

Individual sedentary activities and cognitive function in middle-aged and older adults: A systematic review

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Abstract

Background: The influence of lifestyle factors on cognitive health, particularly among middle-aged and older adults, has garnered significant attention in gerontology and cognitive neuroscience. Currently, over 55 million people worldwide are living with dementia, including Alzheimer's disease with nearly 10 million new cases annually. Although sedentary behavior has been associated with cognitive decline, these studies often treat sitting time as a homogeneous entity, without considering the specific nature of sedentary activities. | **Objective:** This systematic review aims to examine associations of individual sitting activities with cognitive function in middle-aged and older adults. | **Methods:** Data were searched using eight electronic databases (EMBASE, Web of Science, PsycINFO, CINAHL, Medline, SPORTDiscus, PubMed, and Scopus) from inception to September 2024. Qualitative studies, reviews, conference abstracts, theses, and book chapters were excluded. The methodological quality of included studies was assessed using the QualSyst tool. | **Results:** A total of 85 studies (n = 1,575,657) were included in this review. Of the 43 studies examining television viewing, 28 (65%) reported a negative association between prolonged television viewing and cognitive function. Conversely, among the 58 studies that included active sedentary activities, only five (8.6%) reported negative associations. | **Conclusions:** The cognitive effects of sedentary behavior depend on the type of activity performed. Promoting cognitively engaging sedentary activities may support healthy cognitive aging, while excessive passive behaviors may increase risk of cognitive decline and Alzheimer's disease. Future research should focus on further clarifying the mechanisms behind these associations and explore interventions to enhance cognitive health in aging populations.

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